EXTERN + OUTAMATION
EXTERNSHIP

ADVANCED AI-POWERED

DOCUMENT INSIGHT AND DATA EXTRACTION PROJECT

SPRINT 5 STATUS UPDATE

by Lashawn Fofung, MBA, PMP, CSM, CSPO

SKILLS

- Python
- AI/ML
- Data Scrapping
- Prompt Engineering
- LLMs
- Text Extraction
- Document Intelligence
- Python Automation
- Optical Character Recognition (OCR)
- Retrieval-Augumented Generation (RAG)

Helping businesses improve, automate, and architect critical tasks (such as document handling, customer service, and data management) by saving thousands of hours and boosting accuracy.

Mortgage Documents

SPRINTS

2025

01 AI Document Intelligence
02 Python & Google Colab
03 Python Data Extraction
04 Optimizing OCR
05 Implementing RAG
06 Optimizing RAG Pipelines
07 Blob Processing & Classification
08 Interactive Chatbot
09 AI-Powered Document Automation Platform

PROJECT TIMELINE

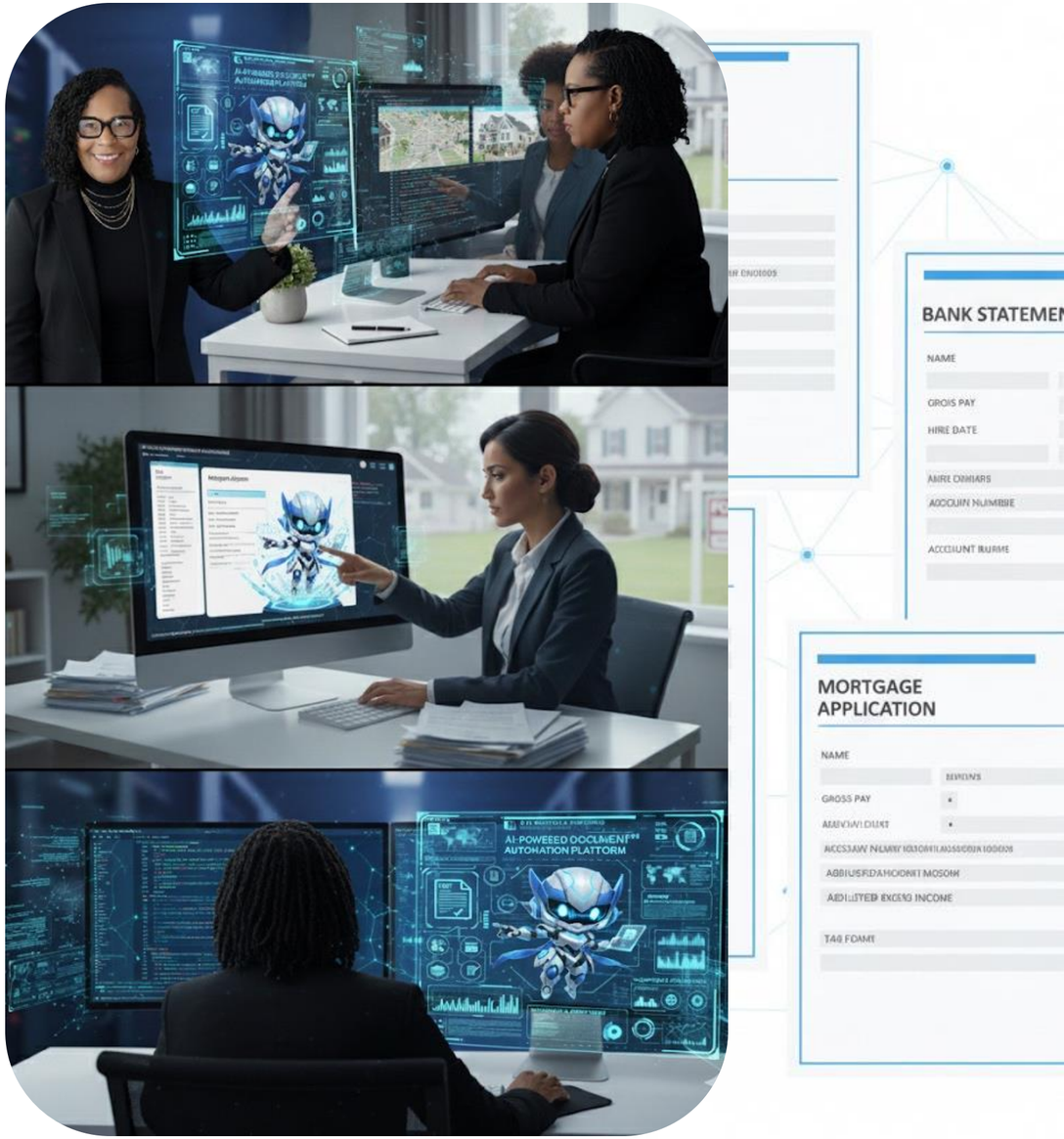
10 WEEKS

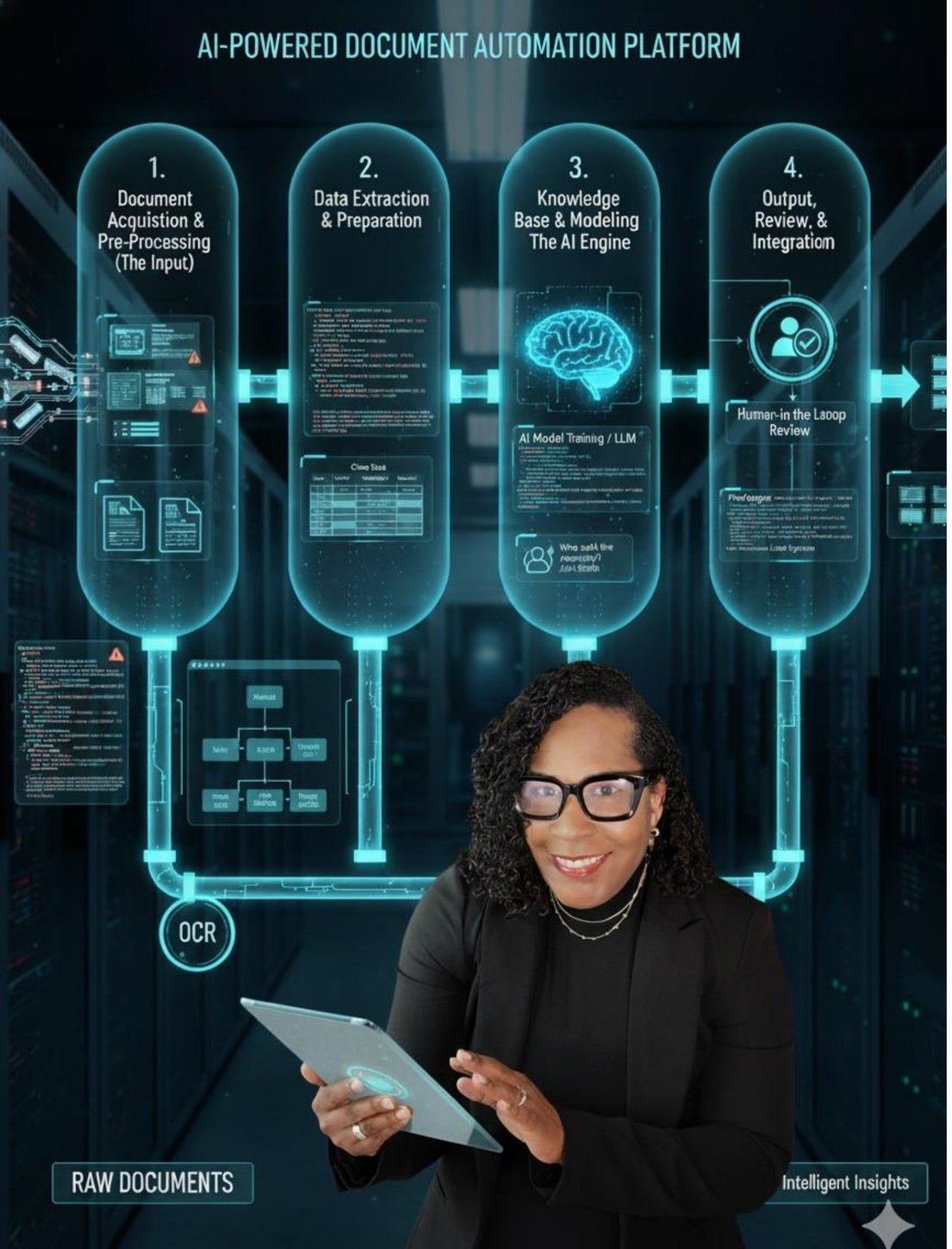
Implementing RAG

Rapid Data Retrieval with LlamaIndex

Background: 10-Week Project

The Outamation Advanced AI-Powered Document Insights and Data Extraction Externship is a comprehensive 10-week program. Its core goal and purpose are to master the required skills to create an AI-Powered Document Automation Platform. The program focuses on bridging foundational technologies (ML, LLMs, NLP, Computer Vision) with high-value industry applications, such as solving complex, document-heavy challenges in mortgage automation, ultimately transforming slow, manual review processes into automated, efficient workflows.





Challenge & Solution: Building The Document Intelligence (AI) Pipeline

- **The Business Challenge:**
  - Automating the highly manual and time-consuming process of mortgage underwriting by handling 200-300 pages of unstructured documents.

- **The Solution:**
  - Designing and implementing a robust, AI-Powered Document Automation Platform to transform raw files into structured, intelligent insights.

- **The 4-Step Document Intelligence Workflow:**
  - Step 1. Document Acquisition & Pre-Processing (The Input)
  - Step 2. Data Extraction & Preparation
  - Step 3. Knowledge Base & Modeling (The AI Engine)
  - Step 4. Output, Review, & Integration (The Results)

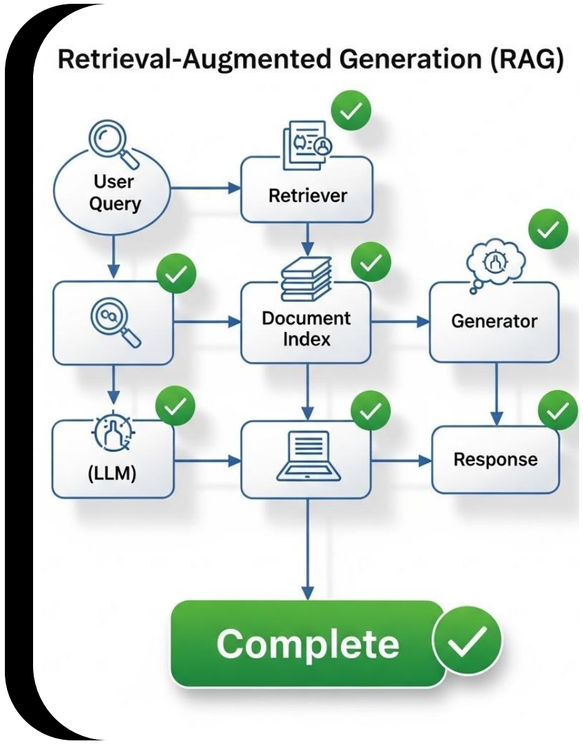
SPRINT 5 FOCUS

Steps 3: Knowledge Base & Modeling (The AI Engine)

Sprint 5 focus was the AI Engine (Step 3). I successfully transformed the clean data into a searchable Knowledge Base, architected the RAG pipeline for precision with high-performance models, and deployed a memory-aware, conversational AI assistant using Gemini 2.5 Flash.

Overall Status: Sprint 5 Complete

The core objective of the project—to build a functional, interactive Retrieval-Augmented Generation (RAG) Chatbot capable of analyzing semi-structured PDF documents and providing grounded, conversational answers—has been successfully completed and demonstrated. The pipeline is fully configured and operational in the Colab environment.



Key Accomplishments

The sprint focused on optimizing the pipeline components for speed, accuracy, and conversational flow, resulting in the following key achievements:

1. Robust PDF Parsing Integration

1. RAG Pipeline Configuration

1. Interactive Chat Engine Development

1. Secure Environment Setup

Robust PDF Parsing

- Successfully integrated and customized the PyMuPDF (fitz) parser
- Handle the visual complexity of the Lender Fees Worksheet
- Ensured accurate text extraction from tables and mixed content
- Critical step for high-quality RAG

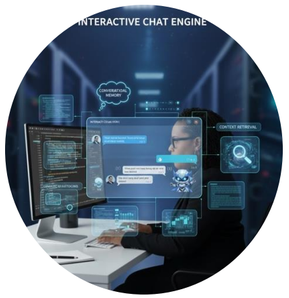
RAG Pipeline Configuration

- Fully configured the LlamaIndex framework, defining the models and chunking strategy.
- LLM Selection: Implemented Gemini 2.5 Flash for high-speed, relevant answer synthesis.
- Embedding Model: Integrated the efficient HuggingFace MiniLM-L6-v2 model for fast, accurate semantic indexing.
- Chunking Strategy: Optimized chunk size (512 tokens with 20 overlap) to maximize context capture for financial facts.

Secure Environment Setup

- Established secure loading of the GEMINI_API_KEY and configured the Hugging Face environment variable for future scalability

Interactive Chat Engine Development



- Upgraded the system from a simple query engine to a ChatEngine
- Enabled the chatbot to maintain memory and context for multi-turn conversations (e.g., answering follow-up questions)

Key Metrics & Progress

The final system demonstrated significant progress against the goals of accuracy and usability.

Metric	Goal	Result	Impact
Data Grounding	100% of generated facts must be present in the source PDF.	Achieved in all demo cases.	High Accuracy. Eliminates AI hallucinations and provides verifiable answers.
Conversational Ability	Successfully handle 2+ turn conversations with follow-up questions.	Complete. The ChatEngine successfully condensed new queries with history.	Enhances user experience by allowing natural, continuous dialogue with the document.
Latency / Efficiency	Use efficient local and preview models to minimize retrieval time.	Optimized. Use of local MiniLM and fast Gemini 2.5 Flash resulted in sub-second response times.	Demonstrates a cost-effective and highly responsive solution for production.
Code Documentation	Comprehensive comments and modular structure for easy hand-off.	Complete. All custom functions and configuration blocks are documented.	Ready for further development and feature expansion.

The Problem: Manual Analysist

Financial documents like Lender’s Fees Worksheet are dense, semi-structured, and error-prone when analyzed manually.

- Time-consuming data extraction
- Risk of missing specifine line items
- Difficulty in calculating cross-referenced totals
- Static keyword search fails on complex intent

The RAG Solution

Retrieval-Augumented Generation for grounded, instant answers.

High-Performance Tech Stack

LlamaIndex

Orchestrates the entire RAG pipeline, managing indexing and retrieval logic.

Gemini 2.5 Flash

The high-speed LLM brain used for reasoning and synthesizing answers.

PyMuPDF

Parsing tool to extract clean text from complex PDF layouts.

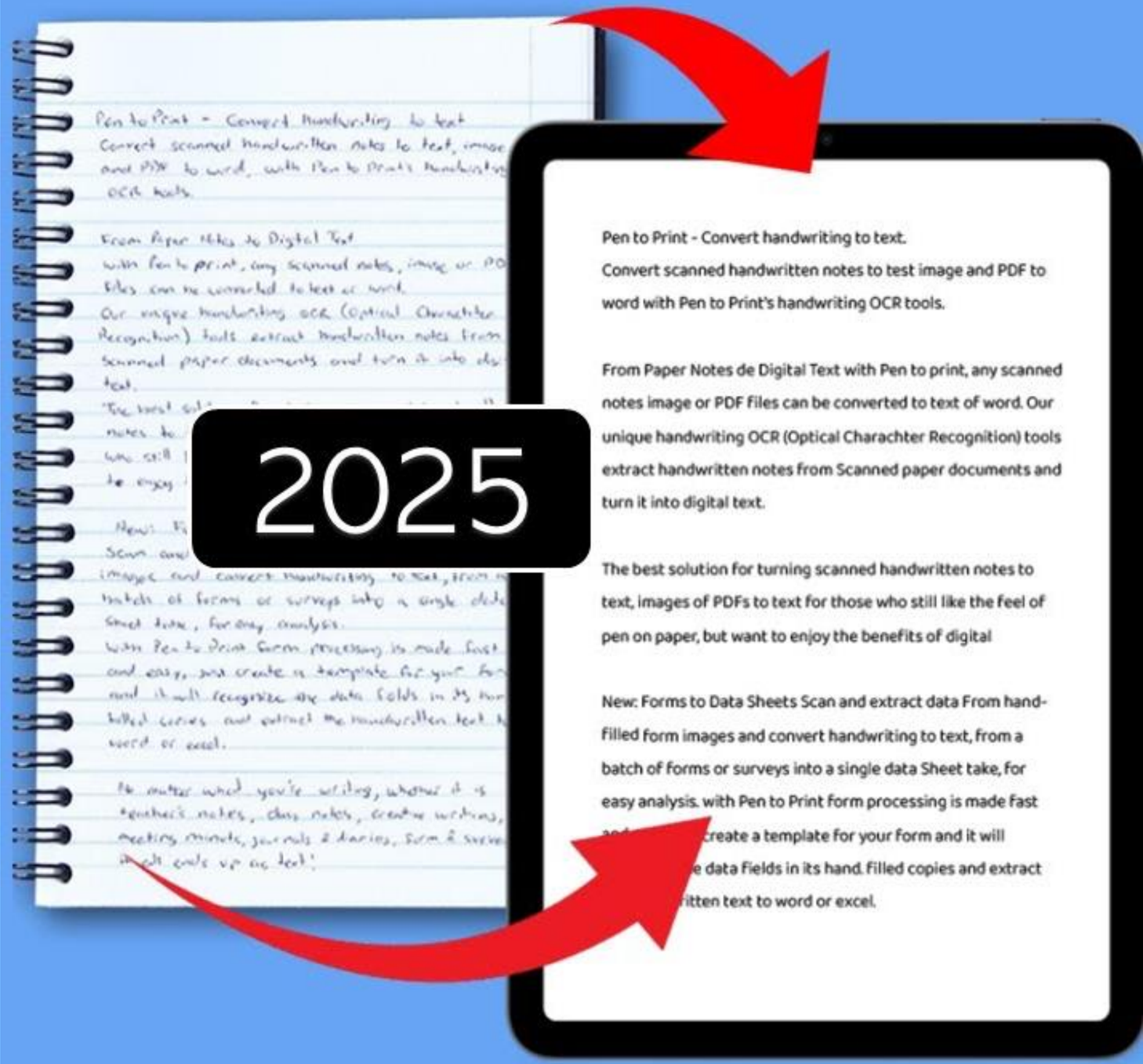
MiniLM-L6-v2

HuggingFace model for efficient, low-latency vector embeddings.

Ingestion & Parsing

Standard PDF loaders struggle with tables. I used PyMuPDF (Fitz) to handle the visual complexity of the “Lender’s Fees Worksheet.”

This ensures the extracted structured data, such as fees, dates, amounts, are cleanly before it reaches the AI model.

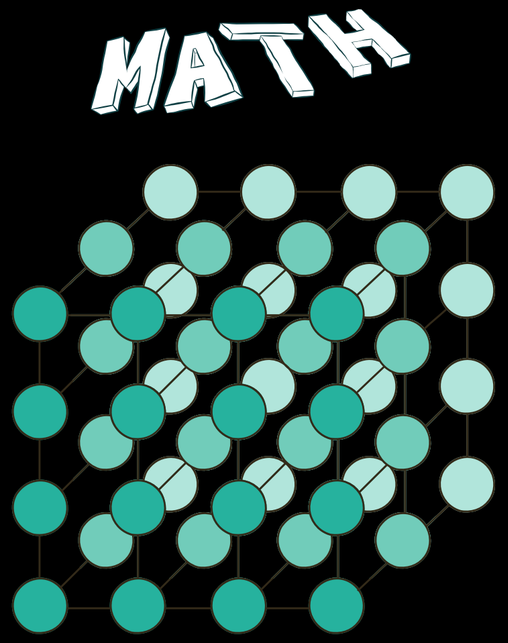


Embedding & Chunking

Converting Text to Math

Breaks the document into 512-token chunks with a 20-token overlap to maintain context boundaries.

The MiniLM-L6-v2 model then converts these chunks into dense vector embeddings, creating a mathematical representation of the document’s meaning.



Business Impact

Efficiency & Accuracy

- **Instant Analysis:**
  - Reduced review time from minutes to seconds.
- **Grounded Answers:**
  - Every response is called from the source document.
- **Scalable:**
  - Architecture supports any document type (Legal, Medical, HR).

DEMOs

Simple Chatbot

Optimized RAG Pipeline with An Interactive RAG Chatbot

CoLab Notebook

- Simple Chatbot with LlamaIndex: HERE
- Optimized RAG Pipeline with An Interactive RAG Chatbot for Document Retrieval: HERE

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SIMPLE CHATBOT

with LlamaIndex

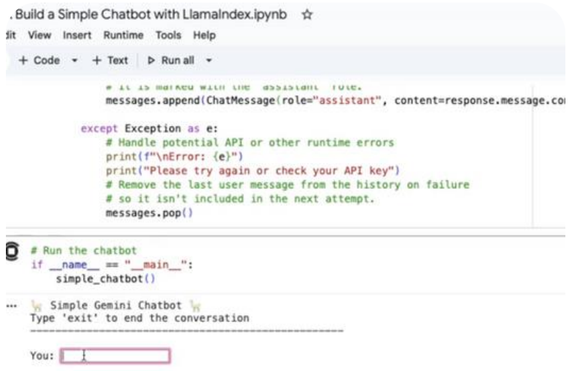
DEMO

HTTPS://YOUTU.BE/FFB6GZT8RFO

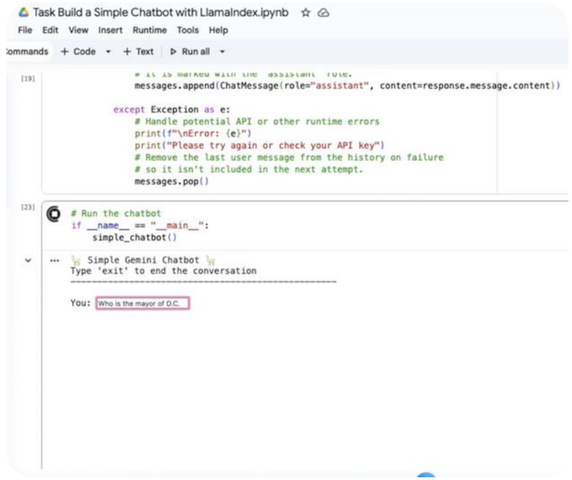
by Lashawn Fofung, MBA, PMP, CSM, CSPO

Demo Chatbot

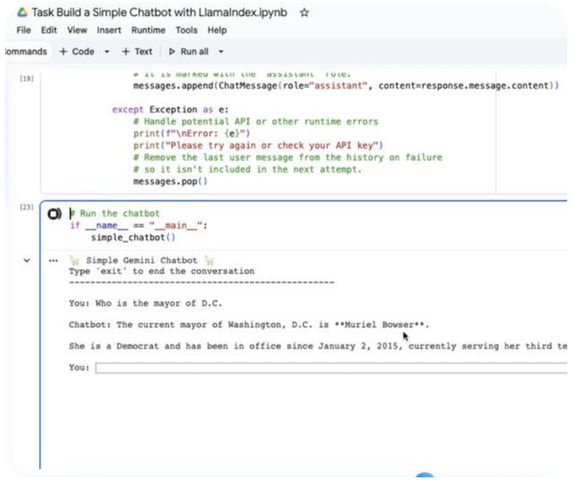
Chatbot



User Input



Response Output



This demonstration showcases a simple, conversational chatbot utilizing a Large Language Model (LLM) for response generation. The goal is to illustrate the fundamental capability of the LLM to process natural language queries and provide answers based only on its extensive pre-trained foundational knowledge, without relying on any external document ingestion or indexing.

Key Components Illustrated

1. **Input**: A standard web input box for user queries.
2. **The LLM:** The core generative model (e.g., Gemini, GPT) that handles the processing and response generation.
3. **Prompt & Inference:** The process where the user query is formatted into a prompt, sent to the model, and the model performs inference to generate the output token-by-token.
4. **Output:** The coherent, conversational response displayed below the input.

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OPTIMIZED RAG PIPELINE

with An Interactive RAG Chatbot

DEMO

HTTPS://YOUTU.BE/E0FKELG2WRU

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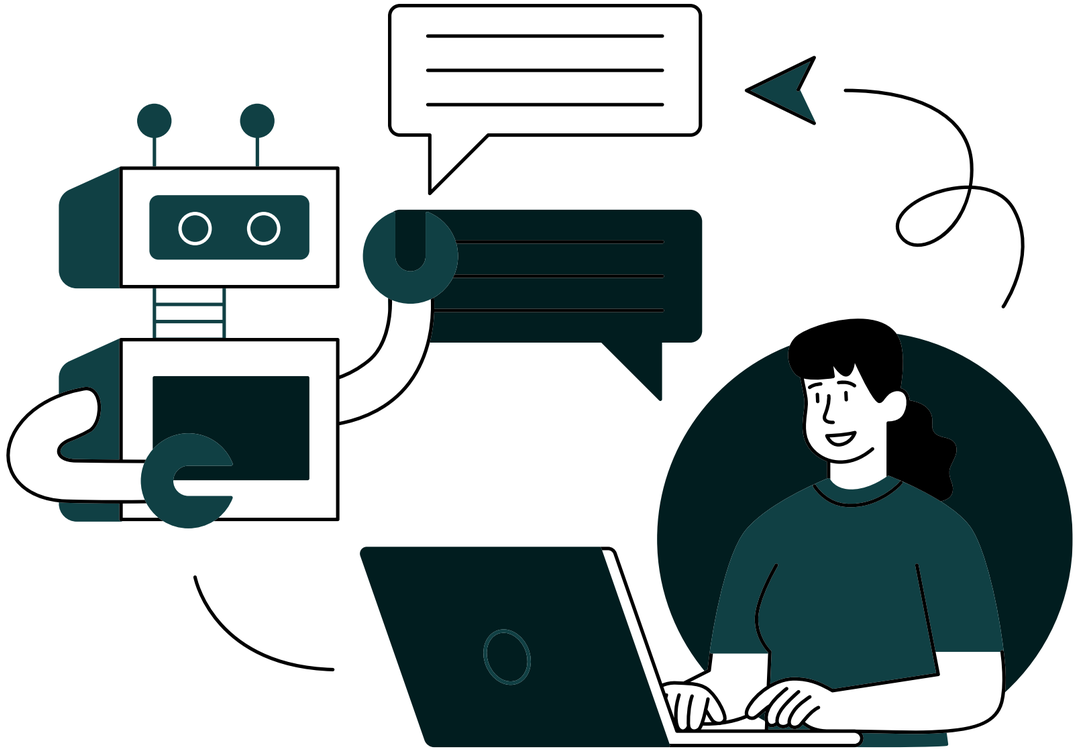
Interactive Chat

From Query to Conversation

The pipeline was upgraded from a simple query engine to a ChatEngine.

This allows the system to maintain conversational memory. A user can ask “What is the fee?” and follow up with “Who pays it?” without repeating the context.



Demo Results: Grounded Truth

3.3 Run the chatbot

[12]
✓ 1m

```
interactive_rag_chatbot()
```

```
--- 💬 Interactive RAG Chatbot 💬 ---
Document: LenderFeesWorksheetNew (1).pdf
------------------------------------------------------------
🟢 Ask Chatbot questions about the document or 🔴 Type  'exit' to end the conversation.
------------------------------------------------------------
You: What is the total estimated monthly payment

Chatbot: The total estimated monthly payment is **$2,308.95**.

This amount is an estimate of your proposed monthly mortgage payment and includes the following components:
*   **Principal & Interest:** $1,869.37
*   **Hazard Insurance:** $39.58
*   **Real Estate Taxes:** $400.00
------------------------------------------------------------
🟢 Ask Chatbot questions about the document or 🔴 Type  'exit' to end the conversation.
------------------------------------------------------------
You: How much of that total is for real estate taxes

Chatbot: Of the total estimated monthly payment of $2,308.95, the amount allocated for **Real Estate Taxes is $400.00**.
------------------------------------------------------------
🟢 Ask Chatbot questions about the document or 🔴 Type  'exit' to end the conversation.
------------------------------------------------------------
You: exit

Chatbot: Goodbye! Feel free to upload a new document next time.
```

THANK YOU

https://github.com/LashawnFofung/AI-Powered-Document-Automation-Platform